INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Third Semester Examination, October-November 2026

Subject Code: BCACC302

Subject Name: Operating Systems

Full Marks: 30 | Set No: 3 | Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. Define Safe Sequence. **2**

2. Define Claim edge. **2**

3. What are the modes of accessing a file? **2**

4. What is virtual memory? State one advantage. **2**

5. Define thrashing in an Operating System. **2**

6. What is a page table? State its purpose. **2**

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. Explain the mechanism of Deadlock Recovery. **5**

8. How disk blocks are allocated for files using Linked allocation method. Explain with a help of an example. **5**

9. Differentiate between internal and external fragmentation with suitable examples. **5**

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. Given memory blocks of sizes 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB, allocate processes of sizes 212 KB, 417 KB, 112 KB, and 426 KB in the given order using First Fit, Best Fit, and Worst Fit. Assume that unused portions of allocated blocks remain available. Which method makes the most effective use of memory for this sequence? **10**

11. Consider the following page reference string: 10

Page reference string

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Number of page frames: 3

Apply the LRU page replacement algorithm and calculate the total number of page faults and page hits.

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Full Marks: 30 | Set No: 1 | Duration: 1 Hour

Part - A
Answer any 5 out of 6 questions
Each question carries 2 marks (2 × 5)

1. Define Deadlock. **2**

2. Explain the concept of mutual exclusion with an example. **2**

3. List and discuss the necessary conditions for deadlock. **2**

4. What is memory management in an Operating System? **2**

5. What is the role of the Memory Management Unit (MMU)? **2**

6. Define demand paging. **2**

Part - B
Answer any 2 out of 3 questions
Each question carries 5 marks (5 × 2)

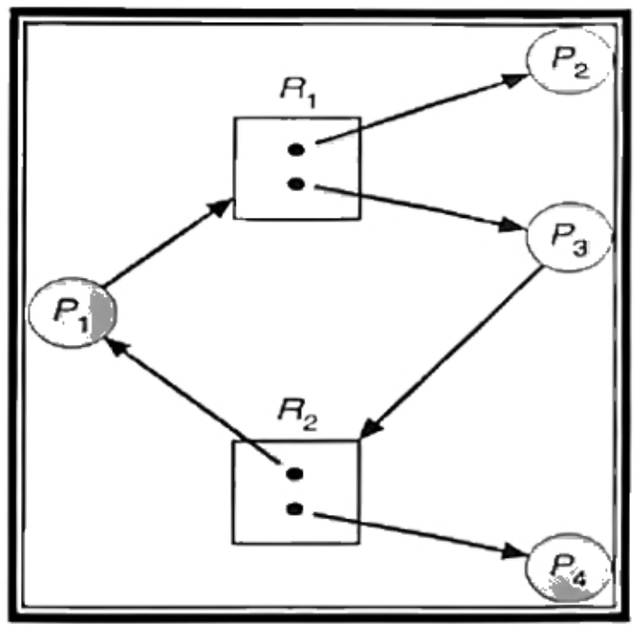
7. Explain the Banker's algorithm for deadlock avoidance with an example. **5**

8. Explain First Fit, Best Fit, and Worst Fit memory allocation techniques with examples. **5**

9. Differentiate between fixed partitioning and dynamic partitioning with suitable examples. **5**

Part - C
Answer any 1 out of 2 questions
Each question carries 10 marks (10 × 1)

10. Explain from the Resource Allocation Graph whether there is a cycle and a Deadlock or no cycle no deadlock or with cycle but no deadlock. **10**



11. Consider the disk queue 98, 183, 122, 37, 14, 124, 65, 67 and apply the FCFS and SSTF Algorithm to find the total head movement of the cylinders, where the head is initially at cylinder 53. The cylinders are numbered ranging from 0 to 199. **10**

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Subject Name: Operating Systems

Full Marks: 30 | Set No: 2 | Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. What is a Translation Lookaside Buffer (TLB)? **2**

2. Define contiguous memory allocation. **2**

3. What is Logical address and Physical address? **2**

4. Describe Resource Allocation Graph. **2**

5. Define Single-Level Directory. **2**

6. What are Sectors and Tracks? **2**

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. How disk blocks are allocated for files using Contiguous allocation method. Explain with a help of an example. **5**

8. Discuss about Hard Link and Soft Link. **5**

9. Explain the concept of segmentation. Differentiate between paging and segmentation. **5**

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. Create the Safe Sequence of the following Snapshot at time T_0: for 5 processes P_0 through P_4 having 3 resource types A (10 instances), B (5instances, and C (7 instances). **10**

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

11. Consider the following page reference string: **10**

Page reference string

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

Number of page frames: 3

Apply the FIFO page replacement algorithm and determine the total number of page faults and page hits.